#1 Let $x_1, \ldots, x_n$ be sample of real valued data

Let $\bar{x}_n = \frac{1}{n} \sum_{j=1}^{n} x_j$ denotes the sample mean.

a) show that $\bar{x}_n = \arg\min_{a \in \mathbb{R}} \sum_{j=1}^{n} (x_j - a)^2$

$$f(a) = \sum_{j=1}^{n} (x_j - a)^2 \xrightarrow[\text{set equal to zero}]{\text{take derivative on both sides}}$$

$$f'(a) = \sum_{j=1}^{n} 2(x_j - a)(-1) = -2 \sum_{j=1}^{n} (x_j - a) = 0$$

$$\sum_{j=1}^{n} x_j - na = 0 \Rightarrow a = \frac{1}{n} \sum_{j=1}^{n} x_j = \bar{x}_n$$

since $f''(a) = 2n > 0 \rightarrow$ this is global minimum

b) $x_1, \dots, x_n$ of d dimensional feature vectors $d \geq 2$ and $\bar{x}_n$ is sample mean. Show

$$\bar{x}_n = \arg\min_{a \in \mathbb{R}^d} \sum_{j=1}^{n} \|x_j - a\|^2$$

Methodolgy: let $A = (a_{ij})_{i,j=1,\dots,n}$ be an $(n \times n)$ matrix. Let $\|A\|_F^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2$

Show: $\|A\|_F^2 = \text{trace}(A^T A)$.

Explain how this quantity can be expressed thru the eigenval of A.

Show that $\|A\|_F^2 = \text{trace}(A^T A)$

$\rightarrow$ compute entry (i,i) for $A^T A$:

$$(A^T A)_{ii} = \sum_{k=1}^{n} (A^T)_{ik} A_{ki} = \sum_{k=1}^{n} a_{ki} \cdot a_{ki} = \sum_{k=1}^{n} a_{ki}^2$$

$$\Rightarrow \text{trace}(A^T A)_{ii} = \sum_{i=1}^{n} \sum_{k=1}^{n} a_{ki}^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij}^2 = \|A\|_F^2$$

Since $A^T A$ is symmetric positive semidefinit

$\rightarrow$ it has real non-neg eigenvalues $\lambda_1, \dots, \lambda_n$

$\rightarrow$ trace $= \sum_{i=1}^{n} \lambda_i$

$$\|A\|_F^2 = \text{trace}(A^T A) = \sum_{i=1}^{n} \lambda_i$$

$\rightarrow$ where $\lambda_i \geq 0$ are the eigenvalues of $A^T A$ $\rightarrow$ the squared singular values of A

a matrix M is PSD if for every vector x: $x^T M x \geq 0$